



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SUNDHARA AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

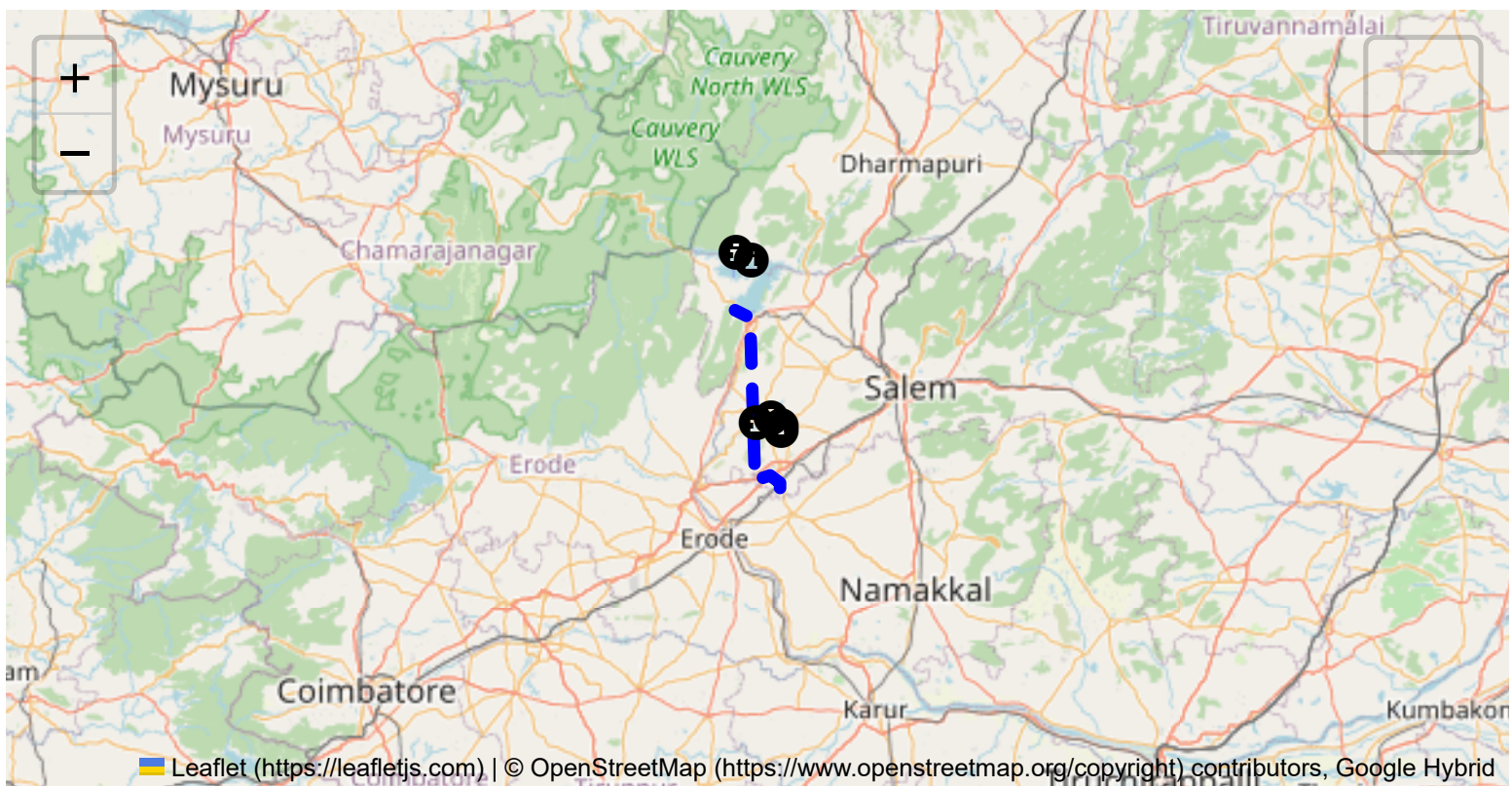
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

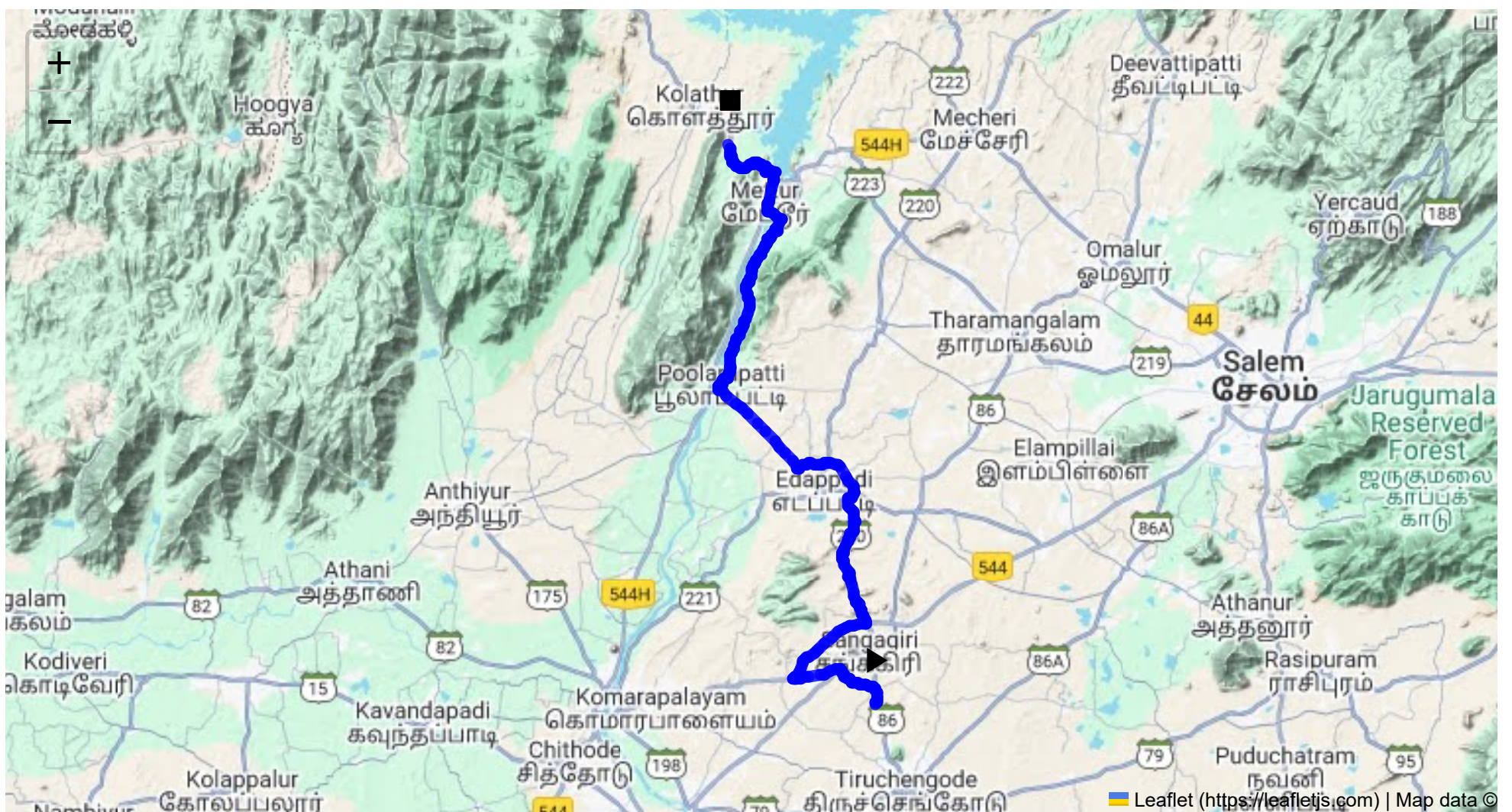
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 66.11 km
Estimated Duration: 1.6 hours
Adjusted Duration (Heavy Vehicle): 2.0 hours
Start: (11.4381, 77.8734)
End: (11.821142, 77.771315)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route spans approximately 66.11 kilometers through Tamil Nadu, India. It begins in Sangagiri, with checkpoints at Sankari RS, Sankari West, Padaiveedu, and Mettur, before terminating at Mulakkadu. The journey traverses a mix of rural, semi-urban, and industrial areas.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The region typically experiences tropical weather, with a hot climate during summer months (March to June) and heavy monsoon rains from October to December. Monsoon rains may cause slippery roads, reduced visibility, and occasional flooding, especially in low-lying areas along the route.

3. Analysis of Traffic Patterns

Traffic congestion is expected during peak hours (8:00-10:00 AM and 5:00-7:00 PM) particularly near the industrial zones and at the intersection points like Sankari RS and Bhavani Main Road. Expect slower movement due to local traffic, especially in Sankari West and Mettur.

4. Assessment of Road Quality and Infrastructure

The roads along this route are generally two-lane highways with regular maintenance. However, certain stretches might have potholes or uneven surfaces, especially after the monsoon season. Signage is adequate, but night driving requires caution due to limited street lighting in rural sections.

5. Suggestions for Alternative Routes for Emergencies

For emergencies, consider taking the NH44 and NH544 routes, which are major highways offering quicker access to emergency services and better road conditions but may increase travel distance.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu regulations mandate specific licenses for transporting hazardous materials, adherence to weight limits, and ensuring the truck is marked with the appropriate placards. Trucks must avoid transit through densely populated areas during peak hours unless necessary.

7. Overview of Historical Incidents

While the route does not have a high incidence rate, minor accidents occasionally occur due to inadequate infrastructure in some sections, particularly during adverse weather. There have been isolated incidents involving heavy vehicles in Mettur owing to industrial activities.

8. Environmental Considerations and Sensitive Areas

Mettur is a notably sensitive area due to its proximity to industries and the Mettur Dam. Routes passing through forested regions near Sangagiri may have wildlife crossings, requiring reduced speeds and heightened awareness.

9. Analysis of Communication Coverage

Communication coverage is generally strong along the entire route, although some potential dead zones may be encountered around rural pockets near Sankari West and Padaiveedu. Essential for truckers to

have two-way radios as a backup.

10. Estimated Emergency Response Times

Expected response times range from 15-30 minutes in urban and semi-urban sections like Mettur, while rural sections might experience longer waits of 30-45 minutes due to distance from emergency facilities.

11. Overall Summary of Risk Assessment

The route from Sangagiri to Mulakkadu presents moderate risks primarily from weather-related hazards and peak hour congestion. Road quality is adequate but requires caution in less developed areas. Regulatory compliance for hazardous materials is critical throughout the journey. Awareness of sensitive environmental zones and maintaining good communication are key safety considerations. Despite its challenges, the route can be managed effectively with proper planning and adherence to safety protocols.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60019, 77.81968	15 KM/Hr
2	Roundabout	High	11.77576, 77.79789	15 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.44900, 77.87406	30 KM/Hr
6	Turn	Medium	11.45357, 77.85789	30 KM/Hr
7	Turn	High	11.45352, 77.85698	15 KM/Hr
8	Turn	Medium	11.45847, 77.85140	30 KM/Hr
9	Turn	Medium	11.45898, 77.85135	30 KM/Hr
10	Turn	High	11.46303, 77.84945	15 KM/Hr
11	Turn	High	11.45542, 77.81725	15 KM/Hr
12	Turn	High	11.45634, 77.81673	15 KM/Hr
13	Blind Spot	Blind Spot	11.49397, 77.86740	10 KM/Hr
14	Turn	Medium	11.49404, 77.86724	30 KM/Hr
15	Turn	Medium	11.50866, 77.86053	30 KM/Hr
16	Turn	Medium	11.52587, 77.85524	30 KM/Hr
17	Turn	Medium	11.54750, 77.85273	30 KM/Hr
18	Turn	Medium	11.55039, 77.85511	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
19	Turn	Medium	11.55481, 77.85639	30 KM/Hr
20	Turn	Medium	11.56411, 77.85969	30 KM/Hr
21	Turn	High	11.57903, 77.85441	15 KM/Hr
22	Turn	Medium	11.58022, 77.85744	30 KM/Hr
23	Turn	High	11.65398, 77.76444	15 KM/Hr
24	Turn	High	11.65608, 77.76465	15 KM/Hr
25	Turn	High	11.65623, 77.76451	15 KM/Hr
26	Turn	High	11.65760, 77.76513	15 KM/Hr
27	Turn	Medium	11.65781, 77.76618	30 KM/Hr
28	Turn	High	11.66103, 77.76757	15 KM/Hr
29	Turn	High	11.66090, 77.76849	15 KM/Hr
30	Turn	Medium	11.66123, 77.76890	30 KM/Hr
31	Turn	High	11.66122, 77.76992	15 KM/Hr
32	Turn	High	11.66386, 77.77083	15 KM/Hr
33	Turn	High	11.66400, 77.77136	15 KM/Hr
34	Turn	High	11.66857, 77.77231	15 KM/Hr
35	Turn	High	11.66873, 77.77191	15 KM/Hr
36	Turn	High	11.66911, 77.77183	15 KM/Hr
37	Turn	Medium	11.66947, 77.77209	30 KM/Hr
38	Turn	Medium	11.72539, 77.78362	30 KM/Hr
39	Turn	Medium	11.72628, 77.78415	30 KM/Hr
40	Turn	High	11.72745, 77.78411	15 KM/Hr
41	Turn	Medium	11.72759, 77.78433	30 KM/Hr
42	Turn	Medium	11.72744, 77.78519	30 KM/Hr
43	Turn	High	11.72785, 77.78582	15 KM/Hr
44	Turn	Medium	11.73941, 77.78811	30 KM/Hr
45	Turn	Medium	11.73971, 77.78838	30 KM/Hr
46	Turn	Medium	11.74000, 77.78938	30 KM/Hr
47	Turn	Medium	11.75274, 77.79710	30 KM/Hr
48	Turn	High	11.75599, 77.79872	15 KM/Hr
49	Turn	Medium	11.75676, 77.79842	30 KM/Hr
50	Turn	High	11.75724, 77.79860	15 KM/Hr
51	Turn	Medium	11.75752, 77.79996	30 KM/Hr
52	Turn	Medium	11.76229, 77.80500	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
53	Turn	High	11.76312, 77.80533	15 KM/Hr
54	Turn	Medium	11.76347, 77.80512	30 KM/Hr
55	Turn	High	11.76402, 77.80425	15 KM/Hr
56	Turn	Medium	11.76436, 77.80420	30 KM/Hr
57	Turn	Medium	11.76990, 77.80761	30 KM/Hr
58	Turn	High	11.77028, 77.80821	15 KM/Hr
59	Turn	High	11.77045, 77.80821	15 KM/Hr
60	Turn	High	11.79781, 77.80274	15 KM/Hr
61	Turn	High	11.79815, 77.80206	15 KM/Hr
62	Turn	Medium	11.80241, 77.80336	30 KM/Hr
63	Turn	High	11.80269, 77.80331	15 KM/Hr
64	Turn	Medium	11.80472, 77.80023	30 KM/Hr
65	Turn	Medium	11.80475, 77.79923	30 KM/Hr
66	Blind Spot	Blind Spot	11.80338, 77.79756	10 KM/Hr
67	Turn	Medium	11.80364, 77.79741	30 KM/Hr
68	Turn	High	11.80665, 77.79714	15 KM/Hr
69	Turn	Medium	11.80793, 77.79256	30 KM/Hr
70	Turn	Medium	11.80877, 77.79170	30 KM/Hr
71	Turn	Medium	11.80856, 77.78620	30 KM/Hr
72	Turn	Medium	11.80508, 77.78132	30 KM/Hr
73	Turn	Medium	11.81047, 77.77325	30 KM/Hr
74	Turn	Medium	11.81463, 77.77229	30 KM/Hr
75	Turn	Medium	11.81958, 77.77226	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	S.K.A Hospital	11.7816111, 77.797688	30 km/h	Medium
3	hospital	Sathya Hospital	11.7864589, 77.8004201	30 km/h	Medium
4	clinic	Saraswathi Poly Clinic	11.7905491, 77.8007345	30 km/h	Medium
6	clinic	Duraisamy Eye Clinic	11.7942636, 77.7995843	30 km/h	Medium
7	clinic	Shiva Siddha Clinic	11.794237, 77.7995531	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
8	hospital	Govt Hospital Mettur	11.7959119, 77.7987959	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
1	college	Govt. Arts College, Mettur	11.7756806, 77.7958483	30 km/h	Medium
5	school	Govt. Hr. Sec. School	11.792309, 77.8006337	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58300, 77.85813



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.60019, 77.81968



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.77576, 77.79789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45847, 77.85140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45898, 77.85135



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45634, 77.81673



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49397, 77.86740



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49404, 77.86724



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50866, 77.86053



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52587, 77.85524



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.54750, 77.85273



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.55039, 77.85511



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55481, 77.85639



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56411, 77.85969



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.57903, 77.85441



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58022, 77.85744



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.65398, 77.76444



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.65608, 77.76465



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.65623, 77.76451



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.65760, 77.76513



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.65781, 77.76618



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66103, 77.76757



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66090, 77.76849



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66123, 77.76890



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66122, 77.76992



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66386, 77.77083



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66400, 77.77136



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66857, 77.77231



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66873, 77.77191



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66911, 77.77183



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.66947, 77.77209



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72539, 77.78362



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72628, 77.78415



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.72745, 77.78411



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72759, 77.78433



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72744, 77.78519



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.72785, 77.78582



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73941, 77.78811



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73971, 77.78838



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74000, 77.78938



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.75274, 77.79710



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.75599, 77.79872



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.75676, 77.79842



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.75724, 77.79860



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.75752, 77.79996



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.76229, 77.80500



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.76312, 77.80533



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.76347, 77.80512



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.76402, 77.80425



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.76436, 77.80420



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.76990, 77.80761



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.77028, 77.80821



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.77045, 77.80821



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.79781, 77.80274



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.79815, 77.80206



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.80241, 77.80336



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.80269, 77.80331



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.80472, 77.80023



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.80475, 77.79923



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.80338, 77.79756



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.80364, 77.79741



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.80665, 77.79714



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.80793, 77.79256



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.80877, 77.79170



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.80856, 77.78620



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.80508, 77.78132



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.81047, 77.77325



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.81463, 77.77229



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.81958, 77.77226
